

Particle Track Reconstruction as a Spin-Glass Optimization Problem

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Part I

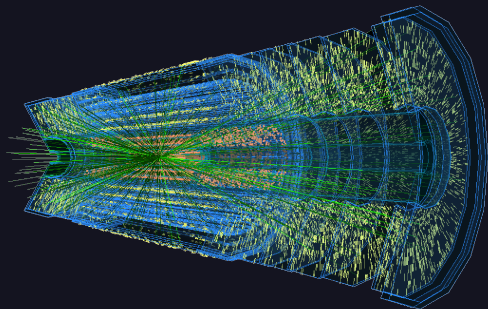
Design & Implementation

Giovanni Guidarini

The Combinatorial Explosion of the LHC

- ▶ LHC collisions create many charged particles.
- ▶ Detectors observe discrete **hits**, not trajectories.
- ▶ Goal: recover **tracks** from layer-wise hits.
- ▶ Search space grows as N^L .

Simplified 2D setup to isolate the optimization challenge.



Track Finding as Energy Minimization

Key idea: convert combinatorics into energy minimization.

Candidate segment $\leftrightarrow$ binary variable:

$$x_i \in \{0, 1\}, \quad x_i = 1 \Rightarrow \text{segment selected}$$

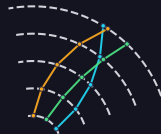
Hamiltonian $\mathcal{H}(\mathbf{x})$ encodes:

- ▶ **Rewards:** smooth, consistent chains
- ▶ **Penalties:** forks, merges, bad geometry

Track finding: $\mathbf{x}^* = \arg \min \mathcal{H}(\mathbf{x})$

Track Finding as Energy Minimization

High energy / Unphysical Low energy / True tracks



The Synthetic Detector

Physics Engine Parameters

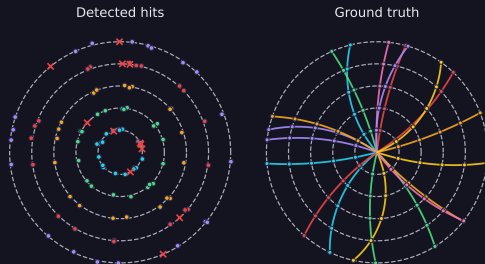
Detector 5 concentric layers ($r = 1, \dots, 5$)

Event size 20 particles $\Rightarrow$ 100 true hits

Motion model circular arcs (uniform B -field)

Noise model Gaussian jitter + Poisson fake hits ($\mu = 2/\text{layer}$)

Particles sampled uniformly in ϕ , curvature r and direction (Clockwise/Counter-Clockwise).



The Hamiltonian

Three contributions encoding the physics of track formation

$$\mathcal{H}(\mathbf{x}) = \underbrace{-\frac{1}{2} \sum_{i,j} J_{ij} x_i x_j}_{\text{pairwise interactions}} - \underbrace{\sum_i h_i x_i}_{\text{local field}} - \underbrace{\sum_{(j,i,k) \in \tau} E_{\text{triplet}}(j,i,k) x_j x_i x_k}_{\text{three-body curvature}}$$

Pairwise J_{ij}

Rewards aligned concatenations.
Penalizes forks, merges, sharp bends.

Local field h_i

Penalizes long segments and non-radial inner-layer starts.

Triplet E_{triplet}

Enforces curvature consistency across 3 consecutive segments.

Pairwise Interactions J_{ij}

Reward alignment, penalize conflicts

Alignment score: $a_{ij} = \frac{\mathbf{s}_i \cdot \mathbf{s}_j}{\|\mathbf{s}_i\| \|\mathbf{s}_j\|}$

Concatenated segments

If $a_{ij} > \cos(\theta_{\max})$:

$$J_{ij} \leftarrow \begin{cases} a_{ij} + \beta_c & \text{if } |\Delta\phi_{ij}| < \theta_{\text{tol}} \\ a_{ij} - \lambda_c & \text{otherwise} \end{cases}$$

Else (sharp turn): $J_{ij} \leftarrow -\lambda_\theta$

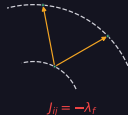
Conflicts

Fork (same start hit): $J_{ij} \leftarrow -\lambda_f$

Merge (same end hit): $J_{ij} \leftarrow -\lambda_m$

All other pairs: $J_{ij} = 0$

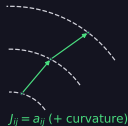
Fork



Merge



Alignment



Local Field & Triplet Interaction

Local field h_i

Length penalty (all segments):

$$h_i \leftarrow -\lambda_\ell \ell_i$$

Radial penalty ($0 \rightarrow 1$ only): if direction deviates from radial by $> \theta_{\text{rad}}$,

$$h_i \leftarrow -\lambda_r$$

Both terms enter $\mathcal{H}$ as $-\sum h_i x_i$, so negative h_i penalizes selection.

Triplet term $E_{\text{triplet}}(j, i, k)$

For three consecutive segments, compare turning angles $\kappa(j, i)$ and $\kappa(i, k)$:

$$E_{\text{triplet}} \leftarrow \begin{cases} +\beta_c & \text{if } |\kappa(j, i) - \kappa(i, k)| < \theta_{\text{tol}} \\ -\lambda_c & \text{otherwise} \end{cases}$$

Rewards near-constant curvature (circular arcs).



Simulated Annealing

Navigating the energy landscape with Metropolis-Hastings

Metropolis step: flip a random spin and accept with

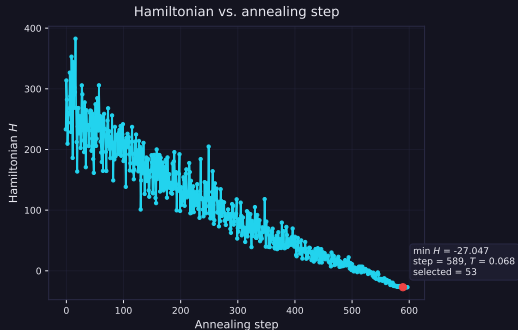
$$\mathbb{P}_{\text{accept}} = \min\left\{1, \exp\left(-\frac{\Delta\mathcal{H}}{T}\right)\right\}$$

Cooling schedule (linear):

$$T(n) = T_{\max} - \frac{n}{N_{\text{iter}} - 1} (T_{\max} - T_{\min})$$

Implementation: solver in **C++**, analysis in **Python**, tuning with **Optuna**.

Annealing Hamiltonian Trace



Part II

Optimization & Results

Nicola Mucciaccio

Evaluation Metrics

Segment-level

TPR (Recall): true segments recovered

Precision: selected segments that are true

F1: harmonic mean of precision and recall

$$F_1 = 2 \frac{\text{Prec.} \times \text{Rec.}}{\text{Prec.} + \text{Rec.}}$$

Track-level

Strict efficiency: 4/4 segments correct

Soft efficiency: at least 3/4 correct

Fake rate: selected segments that are fake

Bifurcations: branching/merging conflicts

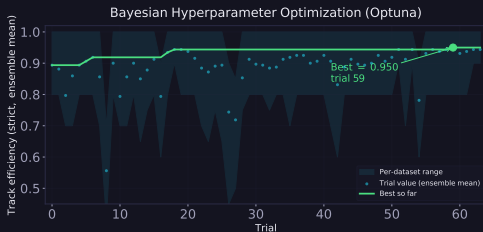
Hyperparameter Tuning

Bayesian optimization with Optuna on the Euler cluster

Two-stage strategy:

1. First optimize on **noise-free** events
(validate model structure first)
2. Then re-optimize on **noisy** events, starting from the clean-case solution

Setup: Tree-structured Parzen Estimator (TPE).
Objective: maximize track efficiency on 16 events.
48–64 parallel workers on Euler.



Results

Averaged over 128 simulated events (20 tracks, 5 layers each)

Optimized on	Test data	TPR	Prec.	Fl	Strict	Soft	Fake
Noise-free	Noise-free	0.934	0.946	0.940	85.3%	93.2%	5.3%
Noise-free	Noisy	0.295	0.465	0.361	1.6%	10.2%	53.5%
Noisy	Noise-free	0.869	0.871	0.870	63.1%	87.0%	13.0%
Noisy	Noisy	0.673	0.728	0.699	30.7%	58.6%	27.2%

Zero bifurcations in all configurations — fork/merge penalties are fully effective.

Noise-Free Performance

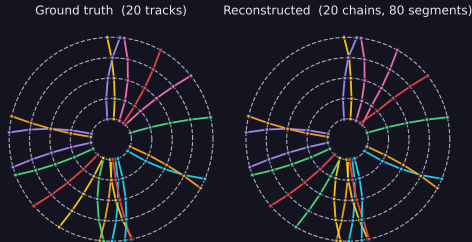
The Hamiltonian captures the essential physics

Clean-data performance:

- ▶ 85.3% strict track efficiency
- ▶ 93.2% soft track efficiency
- ▶ Only 5.3% fake rate

Triplet term is critical: without it, selected chains lose curvature consistency.

Remaining errors occur when tracks are very close on one layer.



The Collapse Under Noise

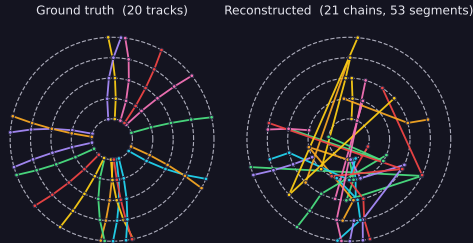
Tight tolerances shatter when exposed to measurement uncertainty

Noise-free tuning on noisy data:

- ▶ Strict efficiency drops 85.3% → 1.6%
- ▶ Fake rate rises to 53.5%
- ▶ Only 53 segments selected (expected 80)

Tight thresholds reject perturbed true segments and admit fake ones.

Re-tuning recovers 30.7% strict / 58.6% soft — genuinely different penalty trade-offs.



The Dominant Failure Mode: Hit Swap

When two tracks pass close on one layer, the model can **swap hits** with similar energy:

$$H_{\text{true}} \approx H_{\text{swapped}}$$

The local constraints (angles, curvature) cannot distinguish the true from the swapped configuration — they are **nearly degenerate** local minima.

Main issue is model expressiveness, not annealing convergence.

Dominant Failure Mode : Hit Swap

True configuration Swapped configuration



Conclusion & Future Directions

Track reconstruction can be framed as a spin-glass optimization. On clean data the method is strong; under noise, local geometry alone is insufficient.

1. Algorithm

Parallel tempering
with replica exchange
to escape shallow
local minima

2. Physics

Four-body terms or
global momentum
conservation to break
the degeneracy

3. Scale

3D extension with
GNN pre-filtering
for realistic $O(10^4)$
hit complexity

Thank you

Questions?